

MODULAR KOSZUL DUALITY AND THE SHADOW TOWER: ALGEBRAICITY, DEPTH CLASSIFICATION, AND THE GRAVITATIONAL COPRODUCT

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ABSTRACT. Let $\mathcal{A}$ be a chirally Koszul vertex algebra with modular characteristic κ , cubic shadow α , and quartic shadow S_4 . We prove four results. First, the weighted shadow generating function $H(t) = \sum_{r \geq 2} r S_r t^r$ satisfies $H(t)^2 = t^4 Q(t)$ where $Q(t) = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4) t^2$ is the *shadow metric*: the tower is algebraic of degree 2, and every S_r for $r \geq 5$ is determined by the three seeds (κ, α, S_4) . Second, the critical discriminant $\Delta = 8\kappa S_4$ partitions the standard landscape into four depth classes: Gaussian ($r_{\max} = 2$), Lie ($r_{\max} = 3$), contact ($r_{\max} = 4$), and mixed ($r_{\max} = \infty$). Third, the genus-2 free energy is $F_2 = 7\kappa/5760$ for single-generator algebras; the first multi-channel computation for $\mathcal{W}_3$ exhibits a cross-channel correction $\delta F_2 = (c+204)/(16c)$, resolving the multi-generator universality question negatively. Fourth, the gravitational coproduct obtained by BRST reduction from $\widehat{\mathfrak{sl}}_2$ to Virasoro is strictly primitive at all degrees; the mechanism applies to all principal Drinfeld–Sokolov reductions.

Remark 0.1 (Relationship to companion papers). The c -independent identity $S_3(\text{Vir}_c) = 2$ is now proved as a theorem; see [?]. The seven-face programme connecting the shadow tower to Gaudin models, Sklyanin brackets, and GZ26 commuting Hamiltonians is developed in [?].

1. INTRODUCTION

1.1. The question. Classical Koszul duality is a phenomenon of graded algebras over a field. A quadratic algebra A determines a dual coalgebra $A^!$; the bar construction $B(A)$ mediates between them; and the Koszul property (bar cohomology concentrated in bar degree 1) is equivalent to the existence of a linear resolution. The theory is the subject of Priddy [?], Beilinson–Ginzburg–Soergel [?], and Loday–Vallette [?].

On an algebraic curve X , the generators of the algebra become sections of a $\mathcal{D}_X$ -module, the quadratic relations become operator product expansions with meromorphic singularities, and the bar differential becomes an integral

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transform whose kernel is the logarithmic 1-form $\eta_{ij} = d \log(z_i - z_j)$ on the Fulton–MacPherson compactification $\overline{C}_n(X)$ of the configuration space. The resulting object, the *chiral bar complex* $\overline{B}^{\text{ch}}(\mathcal{A})$, is a factorization coalgebra on the Ran space $\text{Ran}(X)$ [?, ?, ?].

At genus 0, the classical theory embeds via restriction to the formal disk (the embedding requires specifying a retraction $\mathbb{A}^1 \rightarrow \text{pt}$ and its homotopy transfer; the E_1/E_∞ bar distinction, the Arnol’d algebra of $\text{Conf}_n(\mathbb{A}^1)$, and the Fulton–MacPherson compactifications are already absent over a bare point). At genus $g \geq 1$, the topology of the curve forces further structure: the bar differential acquires curvature $d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$, where $\omega_g = c_1(\mathbb{E})$ is the Hodge class on $\overline{\mathcal{M}}_g$ and the scalar $\kappa(\mathcal{A}) \in \mathbb{Q}(c)$ is an intrinsic invariant of the algebra called the *modular characteristic*. This curvature has no analogue in the classical setting; its geometric origin is the nontriviality of the Hodge bundle over $\overline{\mathcal{M}}_g$.

The present paper addresses a single question: *what invariants does the chiral bar construction produce, beyond the modular characteristic?*

1.2. The answer: the shadow obstruction tower. The bar complex on a curve of genus g with n marked points defines a pronilpotent dg Lie algebra $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ (the modular convolution algebra), whose differential encodes graph sums over stable curves. The boundary-of-boundary relation $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$ implies $D^2 = 0$ on $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$, and the element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ is therefore a Maurer–Cartan element: $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$.

The MC element $\Theta_{\mathcal{A}}$ encodes the full genus expansion. Its finite-order projections $\Theta_{\mathcal{A}}^{\leq r}$, obtained by truncating the degree filtration at level r , form the *shadow obstruction tower*. At degree 2, the tower produces the modular characteristic κ . At degree 3, a cubic shadow α (the structure constant of the transferred L_∞ bracket ℓ_3). At degree 4, a quartic shadow S_4 (the first genuinely nonlinear invariant, measuring the contact term in the four-point collision residue).

The obstruction to truncating $\Theta_{\mathcal{A}}$ at finite degree is measured by classes $o_{r+1}(\mathcal{A}) \in H^2(\mathcal{A}_{r+1,0}^{\text{sh}})$ in the shadow algebra $\mathcal{A}^{\text{sh}} := H_\bullet(\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A}))$. When $o_{r+1} = 0$, the tower stops; when $o_{r+1} \neq 0$, the next shadow coefficient S_{r+1} is forced. The question is how far this process continues and what controls it.

1.3. The shadow metric and algebraicity. A single quadratic polynomial controls the entire shadow obstruction tower (Theorem ??, parts (i)–(ii)).

Restrict to a one-dimensional primary slice L of the cyclic deformation complex, spanned by a single primary field η of definite conformal weight. On L , the MC recursion reduces to a scalar recursion for the shadow coefficients S_r , and the weighted generating function $H(t) = \sum_{r \geq 2} r S_r t^r$ satisfies

$$H(t)^2 = t^4 Q(t), \tag{1.1}$$

where the *shadow metric*

$$Q(t) = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4)t^2 \quad (1.2)$$

is a quadratic polynomial determined by the three seed invariants (κ, α, S_4) . The generating function is algebraic of degree 2: $H(t) = t^2\sqrt{Q(t)}$, and every shadow coefficient S_r for $r \geq 5$ follows from the binomial expansion of $\sqrt{Q}$.

The proof converts the Maurer–Cartan recursion into a convolution identity for the Taylor coefficients of $F(t) = H(t)/t^2$, showing $F^2 = Q$. Orders $m = 0, 1, 2$ reproduce the definitions of κ, α, S_4 ; orders $m \geq 3$ are the recursion.

This is the *intrinsic quartic principle*: three numbers determine an infinite sequence of invariants. The analogue is the Weierstrass equation: two parameters (g_2, g_3) determine all Laurent coefficients of $\wp$, because $\wp$ satisfies an algebraic equation of degree 2 in $\wp'$. Here three parameters (κ, α, S_4) determine all S_r , because H satisfies an algebraic equation of degree 2.

1.4. The four-class classification. Write $Q(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$ for the *Gaussian decomposition*, where $\Delta := 8\kappa S_4$ is the *critical discriminant*. The Gaussian envelope $(2\kappa + 3\alpha t)^2$ is the part of Q visible to the quadratic and cubic shadows alone; Δ measures the irreducible quartic contribution.

The critical discriminant controls a dichotomy.

When $\Delta = 0$, the shadow metric is a perfect square: $Q = (2\kappa + 3\alpha t)^2$, $H(t) = t^2(2\kappa + 3\alpha t)$ is polynomial, and the tower terminates. If $\alpha = 0$, only $S_2 = \kappa$ survives (depth 2); if $\alpha \neq 0$, then S_2 and S_3 survive (depth 3).

When $\Delta \neq 0$, the shadow metric is irreducible: $\sqrt{Q}$ is irrational, every Taylor coefficient beyond degree 1 carries a factor of Δ , and $S_r \neq 0$ for all $r \geq 4$ generically. The tower is infinite.

Every chirally Koszul algebra whose primary-line shadow data has been computed falls into one of four *shadow depth classes* (Theorem ??, part (iii)). The partition is exhaustive for the standard Lie-theoretic landscape (Table ??).

Class	Δ	$r_{\max}$	Prototype	Shadow data
G (Gaussian)	0	2	Heisenberg	$\kappa = k$
L (Lie)	0	3	Affine Kac–Moody	$\kappa, \alpha \neq 0$
C (Contact)	0*	4	$\beta\gamma$ system	$\kappa = 6\lambda^2 - 6\lambda + 1$
M (Mixed)	$\neq 0$	∞	Virasoro, $\mathcal{W}_N$	κ, α, S_4

*Class **C**: $\Delta = 0$ on the primary line; the quartic contact invariant lives on a charged stratum.

Class **G**: the bar complex is classically formal, the tower terminates at κ , and all F_g follow from κ alone. Class **L**: a single cubic correction (the Lie structure constant) appears, after which the Jacobi identity kills the tower. Class **C**: the quartic contact invariant S_4 is nonzero but lives on a separate stratum whose self-bracket exits the complex, terminating the tower one

step later. **Class M:** the tower is infinite, F_g receives corrections at every degree, and the shadow coefficients S_r decay or grow at a rate governed by the *shadow growth rate*

$$\rho(\mathcal{A}) = \sqrt{\frac{9\alpha^2 + 16\kappa S_4}{4\kappa^2}}. \quad (1.3)$$

The growth rate equals the reciprocal of the branch-point modulus. When $\rho < 1$, the tower converges absolutely; when $\rho > 1$, it diverges, but $\sqrt{Q}$ extends analytically past the convergence radius. For Virasoro at central charge c :

$$\rho(\text{Vir}_c) = \frac{1}{c} \sqrt{\frac{180c + 872}{5c + 22}}, \quad (1.4)$$

with a critical central charge $c_\star \approx 6.125$ (the unique positive root of $5c^3 + 22c^2 - 180c - 872 = 0$) separating the convergent and divergent regimes. At the self-dual point $c = 13$ (where the Koszul involution $\text{Vir}_c \mapsto \text{Vir}_{26-c}$ has a fixed point, so Vir_{13} is self-dual under uncurved quadratic Koszul duality): $\rho \approx 0.467$.

Classical Koszul duality over a field embeds into the genus-0, degree-2, $\Delta = 0$ stratum (the Gaussian case where the tower terminates immediately) via restriction to the formal disk; the embedding is not an equivalence, since the configuration-space geometry and the E_1/E_∞ distinction are already absent over a bare point. Everything beyond class **G** is new.

1.5. The shadow connection and Koszul monodromy. The shadow metric equips each primary line with a logarithmic connection

$$\nabla^{\text{sh}} = d - \frac{Q'(t)}{2Q(t)} dt, \quad (1.5)$$

with regular singularities at the zeros of Q and residue $\frac{1}{2}$. The monodromy around each branch point is -1 . This is the *Koszul sign*: the square root $\sqrt{Q}$ changes sign around a simple zero, and the shadow obstruction tower inherits this sign change through the algebraic relation $H = t^2 \sqrt{Q}$.

Under Koszul duality $\mathcal{A} \mapsto \mathcal{A}^!$, the shadow metric transforms by the Verdier involution. For the Virasoro family $\text{Vir}_c^! = \text{Vir}_{26-c}$, the discriminants satisfy

$$\Delta(\text{Vir}_c) + \Delta(\text{Vir}_{26-c}) = \frac{6960}{(5c + 22)(152 - 5c)}, \quad (1.6)$$

a Virasoro-specific identity whose numerator is $6960 = 2^4 \cdot 3 \cdot 5 \cdot 29$. The dual singularities $c = -22/5$ and $c = 152/5$ sum to 26: they are Koszul partners. Other families (affine Kac–Moody, $\mathcal{W}_N$ with $N \geq 3$) produce a different numerator.

1.6. Genus-2 free energies. The shadow obstruction tower produces invariants of moduli spaces through graph sums over stable curves. The genus- g free energy $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$ holds for single-generator (uniform-weight) chirally Koszul algebras at all genera, and unconditionally at genus 1 for all families. Here $\lambda_g^{\text{FP}} = \int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2} \lambda_g$ is the Faber–Pandharipande integral, with $\lambda_g = c_g(\mathbb{E})$ the top Chern class of the Hodge bundle and ψ_1 the cotangent line class.

At genus 2: $\lambda_2^{\text{FP}} = 7/5760$, and $F_2(\mathcal{A}) = \kappa \cdot 7/5760$ for every single-generator chirally Koszul algebra. Verification requires an explicit sum over the seven stable graphs of genus 2 with no marked points (smooth genus-2 curve, figure-eight, banana, dumbbell, theta, tadpole, and chain), weighted by automorphism factors and vertex contributions from the shadow obstruction tower.

For the $\mathcal{W}_3$ algebra, which has two generators T (weight 2) and W (weight 3) producing two primary lines, the genus-2 computation is a multi-channel graph sum: each edge of the stable graph carries a channel label (T or W), each vertex contributes a shadow coefficient on the corresponding primary line, and the propagator $P_T = 2/c$, $P_W = 3/c$ depends on the channel. The per-channel diagonal sum yields $F_2^{\text{scal}}(\mathcal{W}_3) = \kappa \cdot \lambda_2^{\text{FP}} = 7c/6912$, but the full multi-channel graph sum produces an additional cross-channel correction $\delta F_2 = (c + 204)/(16c)$. The multi-generator universality problem is resolved negatively: the cross-channel correction is not cancelled by CohFT R -matrix terms, so $F_g \neq \kappa \cdot \lambda_g^{\text{FP}}$ at genus ≥ 2 for multi-weight algebras. The correct decomposition is $F_g = \kappa \cdot \lambda_g^{\text{FP}} + \delta F_g^{\text{cross}}$.

The Maurer–Cartan equation on $\overline{\mathcal{M}}_{2,0}$ also produces a *planted-forest correction*

$$\delta_{\text{pf}}^{(2,0)} = \frac{S_3(10S_3 - \kappa)}{48}, \quad (1.7)$$

measuring the contribution of codimension-2 boundary strata to the genus-2 amplitude. This correction vanishes for class **G** (where $S_3 = 0$). For classes **C** and **M**, it is generically nonzero. For the Virasoro algebra: $\delta_{\text{pf}}^{(2,0)} = -(c - 40)/48$.

1.7. The gravitational coproduct. The fourth result concerns BRST reduction. Drinfeld–Sokolov (DS) reduction takes affine Kac–Moody data to $\mathcal{W}$ -algebra data via a BRST complex with ghosts; the homological perturbation lemma (HPL) transfers the A_∞ products and coproduct from the affine dg-shifted Yangian to a transferred Yangian on the $\mathcal{W}$ -algebra sector.

We prove the result for $\widehat{\mathfrak{sl}}_2 \rightarrow \text{Vir}$. The ghost-number obstruction (the homotopy h has ghost degree -1 ; the projection $p \otimes p$ kills all ghost-number- (-1) elements) forces the transferred coproduct to be *strictly primitive at all degrees*. The mechanism applies to any principal DS reduction $\widehat{\mathfrak{g}} \rightarrow \mathcal{W}(\mathfrak{g})$, since the ghost-number grading is universal; we state the general case as a

conjecture and prove it for $\widehat{\mathfrak{sl}}_2$:

$$\Delta_z^{\text{Vir}}(T) = \tau_z(T) \otimes 1 + 1 \otimes T, \quad \Delta_{z,n}^{\text{Vir}} = 0 \quad \text{for } n \geq 2.$$

The ghost-number mechanism kills all higher-degree coproduct corrections. The only nontrivial gravitational structure is the r -matrix twist $r^{\text{Vir}}(z) = (c/2)/z^3 + 2T/z$, encoding the braiding of line operators.

The physical content: in Chern–Simons theory, gluons split (the Clebsch–Gordan decomposition makes the coproduct nontrivial). In 3d gravity, gravitons do not split: the coproduct is primitive, and all interactions reside in the twist.

1.8. Relation to existing work. The bar construction for chiral algebras was introduced by Beilinson and Drinfeld [?] and developed in the factorization algebra framework by Francis and Gaitsgory [?] and Costello and Gwilliam [?]. The modular characteristic κ appears implicitly in the work of Feigin and Fuchs on Hodge bundles of conformal blocks, and the formula $\kappa(\text{Vir}_c) = c/2$ can be extracted from Beilinson and Schechtman [?]. The invariants of moduli spaces produced by vertex algebras via conformal blocks are the subject of Faber and Pandharipande [?], Mumford [?], and the extensive literature on cohomological field theories following Kontsevich and Manin [?].

The quartic shadow S_4 , the critical discriminant Δ , the four-class partition, and the algebraicity theorem (Theorem ??) appear to be new. The connection between shadow depth and L_∞ non-formality, while consonant with the general philosophy of Kontsevich [?], has not been established previously in the chiral setting.

The Drinfeld–Sokolov reduction and its homological algebra are classical [?, ?, ?]; the gravitational coproduct primitivity result (Theorem ??) and its connection to the shadow obstruction tower appear to be new.

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The shadow metric, the Riccati algebraicity theorem, and the depth classification Raeez Lorgat

Remark 1.1 (Relationship to companion papers). The Virasoro spectral R-matrix $R(z) = z^{2h} \exp(-(c/4)/z^2)$ and its connection to the Riccati shadow structure are developed in the companion paper *The Virasoro spectral R-matrix* (`virasoro_r_matrix.tex`).

2. THE SHADOW METRIC AND THE RICCATI EQUATION

2.1. The shadow obstruction tower. Let $\mathcal{A}$ be a chirally Koszul vertex algebra with modular cyclic deformation complex $\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A})$ and universal

Maurer–Cartan element $\Theta_{\mathcal{A}} \in \text{MC}(\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A}))$. The complex carries a degree filtration $F^\bullet$ with $F^r = \bigoplus_{s \geq r} \text{Def}_{\text{cyc}}^{(s)}$, and the *shadow obstruction tower* is the sequence of finite-order truncations

$$\Theta_{\mathcal{A}}^{\leq 2}, \Theta_{\mathcal{A}}^{\leq 3}, \Theta_{\mathcal{A}}^{\leq 4}, \dots$$

where $\Theta_{\mathcal{A}}^{\leq r}$ solves the Maurer–Cartan equation in the quotient $\mathfrak{g}_{\mathcal{A}}^{\text{amb}}/F^{r+1}$. The degree- r projection $\text{Sh}_r(\mathcal{A})$ is the r -th *shadow* of $\mathcal{A}$. On a one-dimensional slice of the deformation complex, Sh_r is determined by a single scalar S_r ; the MC equation projected to degree r gives the *all-degree master equation*

$$\nabla_H(\text{Sh}_r) + \mathfrak{o}^{(r)} = 0, \quad (2.1)$$

where ∇_H is the linearized operator and $\mathfrak{o}^{(r)}$ is the obstruction from lower degrees.

2.2. Primary lines and the shadow metric.

Definition 2.1 (Primary line). A *primary line* in $\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A})$ is a one-dimensional subspace $L = \mathbf{k} \cdot \eta$ spanned by a single primary field η of definite conformal weight h_η . For a chiral algebra with generators $\phi_1, \dots, \phi_N$ of weights $h_1, \dots, h_N$, each generator spans a primary line; the full first-order deformation space is $\bigoplus_i L_i$.

Let L be a primary line with coordinate x , and write $\text{Sh}_r|_L = S_r x^r$ for the shadow data restricted to L . Set

$$\kappa := S_2, \quad \alpha := S_3.$$

The scalar κ is the modular characteristic (curvature) of $\mathcal{A}$ on L ; the scalar α is the cubic shadow.

Definition 2.2 (Shadow metric). Define the *weighted initial data* $a_0 := 2\kappa$, $a_1 := 3\alpha$, $a_2 := 4S_4$. The *shadow metric* (shadow quadratic form) on L is the polynomial

$$Q_L(t) := a_0^2 + 2a_0 a_1 t + (a_1^2 + 2a_0 a_2) t^2 = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4) t^2 \in k(c)[t]. \quad (2.2)$$

The *critical discriminant* of L is

$$\Delta := 8\kappa S_4 = a_0 a_2. \quad (2.3)$$

Remark 2.3. The shadow metric Q_L is a quadratic polynomial in t , determined by three OPE invariants: κ , α , S_4 . The term “metric” refers to Q_L ’s role as the quadratic form governing the tower’s geometry (branch points, monodromy, convergence radius), not to a Riemannian metric.

2.3. The recursion on a primary line. On a primary line L , the all-degree master equation (2.1) reduces to a recursion among the scalars S_r .

Proposition 2.4 (Single-line recursion). *On a primary line L with propagator $P := 1/\kappa$, the MC equation at degree $r \geq 5$ is*

$$S_r = -\frac{P}{2r} \sum_{\substack{j+k=r+2 \\ 3 \leq j \leq k}} c_{jk} j k S_j S_k, \quad c_{jk} = \begin{cases} 1 & \text{if } j < k, \\ 1/2 & \text{if } j = k. \end{cases} \quad (2.4)$$

This recursion is well-defined: since $j \geq 3$ and $j+k = r+2$, every term on the right has $k = r+2-j \leq r-1$, so S_r is determined by $\{S_2, S_3, S_4, \dots, S_{r-1}\}$. The degree-4 shadow S_4 is an intrinsic OPE invariant (the quartic contact term in the four-point collision residue), not determined by the recursion; it serves as initial data alongside $\kappa = S_2$ and $\alpha = S_3$.

Proof. The Lie bracket on the cyclic deformation complex, restricted to L , pairs $\text{Sh}_j = S_j x^j$ and $\text{Sh}_k = S_k x^k$ to produce a term of degree $j+k-2$ with coefficient $jk P S_j S_k$. The eigenvalue identity $\nabla_H(x^r) = 2r x^r$ separates the MC equation at degree r into the linear term $2r S_r$ on the left and the obstruction sum on the right, giving the stated recursion. The symmetry factor c_{jk} accounts for the factor $\frac{1}{2}$ in $\frac{1}{2}[\Theta, \Theta]$ when $j = k$. $\square$

2.4. The weighted shadow generating function.

Definition 2.5 (Weighted shadow generating function). The *weighted shadow generating function* on L is

$$H(t) := \sum_{r \geq 2} r S_r t^r. \quad (2.5)$$

Setting $a_n := (n+2) S_{n+2}$ for $n \geq 0$, this becomes $H(t) = \sum_{n \geq 0} a_n t^{n+2}$, with initial values $a_0 = 2\kappa$, $a_1 = 3\alpha$, $a_2 = 4S_4$.

The weighted generating function H packages the recursion (2.4) as a convolution identity. Set $F(t) := H(t)/t^2 = \sum_{n \geq 0} a_n t^n$. The MC recursion forces every Cauchy coefficient of F^2 beyond degree 2 to vanish, so F^2 is a polynomial of degree 2.

3. ALGEBRAICITY AND THE DEPTH CLASSIFICATION

3.1. The Riccati algebraicity theorem.

Theorem 3.1 (Riccati algebraicity). *Let L be a primary line with shadow data S_r , shadow metric Q_L (Definition 2.2), and weighted shadow generating function H (Definition 2.5). Then the all-degree master equation on L is equivalent to the algebraic relation*

$$H(t)^2 = t^4 Q_L(t). \quad (3.1)$$

The shadow generating function is therefore algebraic of degree 2 over $k(\kappa, \alpha, S_4)(t)$:

$$H(t) = t^2 \sqrt{Q_L(t)}. \quad (3.2)$$

Proof. Set $F(t) := H(t)/t^2 = \sum_{n \geq 0} a_n t^n$ where $a_n = (n+2) S_{n+2}$. The claim (3.1) is equivalent to $F(t)^2 = Q_L(t)$. We verify this by computing the Cauchy product $F^2 = \sum_{m \geq 0} (\sum_{i+j=m} a_i a_j) t^m$ and showing it agrees with Q_L at all orders.

Orders 0, 1, 2 (initial data). At $m = 0$: $a_0^2 = (2\kappa)^2 = 4\kappa^2$. At $m = 1$: $2 a_0 a_1 = 2 \cdot 2\kappa \cdot 3\alpha = 12\kappa\alpha$. At $m = 2$: $a_1^2 + 2 a_0 a_2 = 9\alpha^2 + 2 \cdot 2\kappa \cdot 4S_4 = 9\alpha^2 + 16\kappa S_4$. These are exactly the coefficients of $Q_L(t)$ in (2.2).

Orders $m \geq 3$ (the MC recursion forces vanishing). Since Q_L has degree 2 in t , the identity $F^2 = Q_L$ requires

$$\sum_{i+j=m} a_i a_j = 0 \quad \text{for all } m \geq 3. \quad (3.3)$$

We show that this is exactly the MC recursion (2.4) at degree $r = m + 2$.

The sum $\sum_{i+j=m} a_i a_j$ with $a_i = (i+2) S_{i+2}$ and $a_j = (j+2) S_{j+2}$ equals $\sum_{p+q=m+4, p, q \geq 2} p q S_p S_q$, where $p = i+2$, $q = j+2$. Setting $r = m+2$, this becomes $\sum_{p+q=r+2, p, q \geq 2} p q S_p S_q$.

Isolating the terms with $p = 2$ or $q = 2$ (which contribute 2κ from $S_2 = \kappa$):

$$\begin{aligned} \sum_{p+q=r+2, p, q \geq 2} p q S_p S_q &= 2 \cdot 2r \cdot \kappa S_r + \sum_{\substack{p+q=r+2 \\ p, q \geq 3}} p q S_p S_q \\ &= 4r\kappa S_r + 2 \sum_{\substack{j+k=r+2 \\ 3 \leq j \leq k}} c_{jk} j k S_j S_k, \end{aligned}$$

where the factor 2 in front of the restricted sum accounts for $S_p S_q + S_q S_p$ when $p \neq q$, and c_{jk} corrects for the diagonal. Substituting the recursion (2.4):

$$2 \sum_{\substack{j+k=r+2 \\ 3 \leq j \leq k}} c_{jk} j k S_j S_k = 2 \cdot (-2r\kappa S_r) = -4r\kappa S_r.$$

Hence $\sum_{p+q=r+2} p q S_p S_q = 4r\kappa S_r - 4r\kappa S_r = 0$, which is (3.3).

Conversely, the vanishing (3.3) at order $m \geq 3$ implies the recursion (2.4) at $r = m + 2$ by solving for S_r (using $\kappa \neq 0$).

The equivalence of (3.1) with the full MC recursion is therefore established at all degrees. Taking positive square roots (the branch with $F(0) = a_0 = 2\kappa > 0$) gives $H(t) = t^2 F(t) = t^2 \sqrt{Q_L(t)}$. $\square$

Remark 3.2 (Riccati ODE reformulation). Set $U(t) := \sum_{r \geq 3} S_r t^r$ (the non-linear part, removing Sh_2). The algebraic relation (3.1) is equivalent to the Riccati-type first-order ODE

$$2t U'(t) + \frac{P}{2} (U'(t))^2 = R_3 t^3 + R_4 t^4, \quad (3.4)$$

where $R_3 = 6\alpha$ and $R_4 = (9\alpha^2 + 2\Delta)/(2\kappa)$ encode the cubic and quartic inputs. The left-hand side is quadratic in U' : this is the precise sense in which the shadow obstruction tower is algebraic of degree 2.

3.2. Gaussian decomposition.

Corollary 3.3 (Gaussian decomposition). *The shadow metric decomposes as*

$$Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2, \quad (3.5)$$

where $\Delta = 8\kappa S_4$ is the critical discriminant. The first term $(2\kappa + 3\alpha t)^2$ is the Gaussian envelope: the shadow metric of the truncated tower with $S_r = 0$ for $r \geq 4$. The second term $2\Delta t^2$ is the interaction correction, measuring the quartic OPE contribution not captured by the cubic data alone.

Proof. $(2\kappa + 3\alpha t)^2 = 4\kappa^2 + 12\kappa\alpha t + 9\alpha^2 t^2$. Adding $2\Delta t^2 = 16\kappa S_4 t^2$ recovers the coefficient $9\alpha^2 + 16\kappa S_4$ of t^2 in $Q_L(t)$. $\square$

3.3. Intrinsic quartic principle.

Corollary 3.4 (Intrinsic quartic principle). *On a primary line L , the shadow obstruction tower has intrinsic OPE data at degrees 2, 3, 4 only. All shadow coefficients S_r for $r \geq 5$ are determined by (κ, α, S_4) alone, via the closed form*

$$S_r = \frac{1}{r} [t^{r-2}] \sqrt{Q_L(t)} \quad (r \geq 2). \quad (3.6)$$

Proof. $Q_L(t) = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4) t^2$ depends on exactly (κ, α, S_4) . Since $H(t) = t^2 \sqrt{Q_L(t)} = \sum_{r \geq 2} r S_r t^r$ by Theorem 3.1, comparing coefficients gives $r S_r = [t^{r-2}] \sqrt{Q_L}$. $\square$

3.4. The shadow connection.

Proposition 3.5 (Shadow connection). *The shadow metric defines a logarithmic connection on $\mathcal{O}_L$ over $L \setminus \{Q_L = 0\}$:*

$$\nabla^{\text{sh}} = d - \omega, \quad \omega := \frac{Q'_L(t)}{2Q_L(t)} dt = d(\log \sqrt{Q_L}). \quad (3.7)$$

- (i) *The connection is flat: $(\nabla^{\text{sh}})^2 = 0$.*
- (ii) *At each simple zero t_0 of Q_L , the residue of ω is $\frac{1}{2}$.*
- (iii) *The monodromy is $\exp(2\pi i \cdot \frac{1}{2}) = -1$: the standard monodromy of a square root around a simple branch point.*
- (iv) *The flat section with $\Phi(0) = 1$ is $\Phi(t) = \sqrt{Q_L(t)/Q_L(0)} = \sqrt{Q_L(t)/(2\kappa)}$, and the shadow generating function is $H(t) = 2\kappa t^2 \Phi(t)$: the parallel transport of the curvature along the degree parameter, weighted by t^2 .*

Proof. Flatness is automatic: ω is a closed 1-form on a 1-dimensional base. At a simple zero t_0 of Q_L : $Q_L(t) \approx Q'_L(t_0)(t - t_0)$, so $\omega \approx \frac{1}{2(t-t_0)} dt$, giving residue $\frac{1}{2}$. The monodromy exponentiation is immediate. For (iv): $\nabla^{\text{sh}} \Phi = 0$ since $d\Phi = \omega \Phi$ by construction, and the evaluation at $t = 0$ gives $\Phi(0) = \sqrt{Q_L(0)/(2\kappa)} = 2\kappa/(2\kappa) = 1$. $\square$

Remark 3.6. The shadow generating function $H(t) = 2\kappa t^2 \Phi(t)$ is *not* a flat section of ∇^{sh} : the factor t^2 contributes $\nabla^{\text{sh}}(H) \neq 0$. The flat section is $\Phi(t) = \sqrt{Q_L(t)}/(2\kappa)$; the shadow obstruction tower is the Taylor expansion of $t^2 \cdot \Phi(t)$, reflecting the degree offset (the tower starts at degree 2, not degree 0).

3.5. The single-line dichotomy.

Theorem 3.7 (Single-line dichotomy). *Let $L, S_r, \alpha, \Delta, Q_L$ be as above, and define the shadow depth $r_{\max}|_L := \max\{r : S_r \neq 0\}$ (with $r_{\max} = \infty$ if $S_r \neq 0$ for infinitely many r). Then $r_{\max}|_L \in \{2, 3, \infty\}$, classified by the shadow metric:*

- (i) **Class G** ($\Delta = 0, \alpha = 0$). $Q_L = (2\kappa)^2$ is constant. $H(t) = 2\kappa t^2$. Only S_2 survives: $r_{\max} = 2$.
- (ii) **Class L** ($\Delta = 0, \alpha \neq 0$). $Q_L = (2\kappa + 3\alpha t)^2$ is a perfect square. $H(t) = t^2(2\kappa + 3\alpha t)$. S_2 and S_3 survive: $r_{\max} = 3$.
- (iii) **Class M** ($\Delta \neq 0$). Q_L is irreducible over $k(c)$; $\sqrt{Q_L}$ is irrational; $S_r \neq 0$ for all $r \geq 4$ generically: $r_{\max} = \infty$.

For $r \geq 4$,

$$S_r = \Delta \cdot R_r, \quad R_r \in \mathbb{Q}(\alpha, \kappa, S_4). \quad (3.8)$$

Proof. The classification follows from Theorem 3.1 and the Gaussian decomposition (Corollary 3.3).

Case $\Delta = 0$. Then $Q_L = (2\kappa + 3\alpha t)^2$ is a perfect square, so $\sqrt{Q_L} = 2\kappa + 3\alpha t$ is polynomial and $H(t) = t^2(2\kappa + 3\alpha t)$ terminates. If $\alpha = 0$: $H(t) = 2\kappa t^2$, giving $r_{\max} = 2$. If $\alpha \neq 0$: $H(t) = 2\kappa t^2 + 3\alpha t^3$, giving $r_{\max} = 3$.

Case $\Delta \neq 0$. The discriminant of Q_L as a polynomial in t is

$$(12\kappa\alpha)^2 - 4 \cdot 4\kappa^2 \cdot (9\alpha^2 + 16\kappa S_4) = -32\kappa^2\Delta \neq 0,$$

so Q_L is irreducible over $k(c)$. The Taylor expansion of $\sqrt{Q_L}$ around $t = 0$ has nonzero coefficients at all orders generically, hence $r_{\max} = \infty$.

Universal factorization. The binomial expansion of $\sqrt{Q_L}$ around the Gaussian envelope $(2\kappa + 3\alpha t)$ is

$$\sqrt{Q_L} = (2\kappa + 3\alpha t) \sqrt{1 + \frac{2\Delta t^2}{(2\kappa + 3\alpha t)^2}}.$$

Expanding $\sqrt{1+u}$ as $1 + \frac{1}{2}u - \frac{1}{8}u^2 + \dots$ with $u = 2\Delta t^2/(2\kappa + 3\alpha t)^2$, every coefficient of t^n for $n \geq 2$ in $\sqrt{Q_L} - (2\kappa + 3\alpha t)$ carries a factor of Δ . Since $rS_r = [t^{r-2}]\sqrt{Q_L}$ for $r \geq 2$, and the Gaussian part $2\kappa + 3\alpha t$ contributes only at $r \leq 3$, the factorization $S_r = \Delta \cdot R_r$ for $r \geq 4$ follows. $\square$

Remark 3.8 (The contact class). The partition **G/L/C/M** of chirally Koszul algebras includes a fourth class **C** (contact, $r_{\max} = 4$), realized by the $\beta\gamma$ system. This class escapes the single-line dichotomy by *stratum separation*: the quartic contact invariant lives on a charged stratum of $\text{Def}_{\text{cyc}}^{\text{mod}}$ where $\kappa = 0$, and the shadow metric requires $\kappa \neq 0$. On the charged stratum,

the self-bracket maps to a zero-dimensional target by rank-one rigidity, so the quartic pump does not activate and the tower terminates at $r = 4$. The algebraic shadow depth $d_{\text{alg}} \in \{0, 1, 2, \infty\}$ exhausts the four classes: **G** ($d_{\text{alg}} = 0$), **L** ($d_{\text{alg}} = 1$), **C** ($d_{\text{alg}} = 2$), **M** ($d_{\text{alg}} = \infty$).

3.6. The shadow growth rate. For class **M** algebras, the shadow obstruction tower is infinite but governed by a single analytic invariant: the growth rate of the coefficients S_r .

Definition 3.9 (Shadow growth rate). Let L be a primary line of class **M** ($\Delta \neq 0$), and write $Q_L(t) = q_0 + q_1 t + q_2 t^2$ for the shadow metric. The *shadow growth rate* of L is

$$\rho_L := \sqrt{\frac{q_2}{q_0}} = \frac{\sqrt{9\alpha^2 + 2\Delta}}{2|\kappa|}. \quad (3.9)$$

By Vieta's theorem, the zeros $t_0, \bar{t}_0$ of Q_L satisfy $|t_0|^2 = q_0/q_2$, so $\rho_L = 1/|t_0|$: the reciprocal of the branch-point modulus. For classes **G** and **L**, set $\rho := 0$ by convention.

Theorem 3.10 (Shadow asymptotics). *Let L be a primary line of class **M** with shadow growth rate ρ_L .*

(i) Asymptotic formula.

$$S_r \sim A(\mathcal{A}) \rho_L^r r^{-5/2} \cos(r\theta + \varphi) \quad (r \rightarrow \infty), \quad (3.10)$$

where $\theta := -\arg(t_0)$ is the oscillation phase, $A(\mathcal{A}) > 0$, and φ is computable from the branch-point residue of $\sqrt{Q_L}$. The exponent $-5/2$ arises from $S_r = [t^{r-2}] \sqrt{Q_L}/r$ combined with the $n^{-3/2}$ transfer law for square-root singularities.

(ii) Unit circle criterion. $\rho_L = 1$ if and only if $q_0 = q_2$, i.e.,

$$(2\kappa - 3\alpha)(2\kappa + 3\alpha) = 2\Delta. \quad (3.11)$$

Equivalently, the branch points of Q_L lie on the unit circle $|t| = 1$.

(iii) Convergence partition. $\rho_L < 1$: the shadow obstruction tower converges absolutely. $\rho_L > 1$: the shadow obstruction tower diverges, but the algebraic function $\sqrt{Q_L(t)}$ extends analytically past the convergence radius. $\rho_L = 1$: marginal case (the critical surface).

Proof. The function $\sqrt{Q_L(t)}$ has algebraic branch-cut singularities of order $\frac{1}{2}$ at each simple zero t_0 of Q_L . Since Q_L is quadratic with nonzero discriminant ($\Delta \neq 0$ implies $\text{disc}(Q_L) = -32\kappa^2\Delta \neq 0$), the two zeros $t_0, \bar{t}_0$ are distinct with $|t_0| = |\bar{t}_0| = \sqrt{q_0/q_2} = 1/\rho_L$.

The transfer theorem for algebraic singularities [?, Theorem VI.1] gives $[t^n] \sqrt{Q_L} \sim C_0 t_0^{-n} n^{-3/2}$ for each dominant singularity t_0 at distance $R = 1/\rho_L$ from the origin. Both branch points contribute (equal modulus), producing the cosine interference term from $2 \text{Re}(C_0 t_0^{-(r-2)})$. Since $S_r = [t^{r-2}] \sqrt{Q_L}/r$ (Corollary 3.4), the extra factor $1/r$ gives the stated $r^{-5/2}$ exponent.

For (ii): $\rho_L = 1$ iff $q_2/q_0 = 1$. Writing $q_0 = 4\kappa^2$ and $q_2 = 9\alpha^2 + 2\Delta$, the condition $q_0 = q_2$ factors as $4\kappa^2 - 9\alpha^2 = 2\Delta$. Part (iii) is immediate from the definition of the convergence radius $R = 1/\rho_L$. $\square$

3.7. The depth decomposition.

Theorem 3.11 (Depth decomposition). *The shadow depth of a chirally Koszul algebra $\mathcal{A}$ decomposes as*

$$d(\mathcal{A}) = 1 + d_{\text{arith}}(\mathcal{A}) + d_{\text{alg}}(\mathcal{A}), \quad (3.12)$$

where:

- (i) d_{arith} counts the independent holomorphic Hecke eigenforms in the spectral decomposition of the completed shadow partition function $\widehat{Z}_{\mathcal{A}}^c$ on $\mathcal{M}_{1,1}$;
- (ii) $d_{\text{alg}} \in \{0, 1, 2, \infty\}$ is the algebraic depth, determined by the single-line dichotomy (Theorem 3.7) and stratum separation (Remark 3.8):

$$d_{\text{alg}} = \begin{cases} 0 & \text{class } \mathbf{G}, \\ 1 & \text{class } \mathbf{L}, \\ 2 & \text{class } \mathbf{C}, \\ \infty & \text{class } \mathbf{M}; \end{cases}$$

- (iii) depths $d \geq 5$ are purely arithmetic: they arise from d_{arith} , not from the A_∞ structure.

Proof. The algebraic depth d_{alg} is determined by the structure of the cyclic deformation complex and the MC recursion: the single-line dichotomy (Theorem 3.7) gives $d_{\text{alg}} \in \{0, 1, \infty\}$ on each primary line, and stratum separation (Remark 3.8) provides $d_{\text{alg}} = 2$ for the contact class. Since $d_{\text{alg}} \leq 2$ for finite cases ($\mathbf{G}/\mathbf{L}/\mathbf{C}$) and the base term is 1 (the curvature κ at degree 2), the maximum finite depth without arithmetic contributions is $1 + 0 + 2 = 3$; hence all depths $d \geq 5$ require $d_{\text{arith}} \geq 2$.

The arithmetic depth counts independent holomorphic Hecke eigenforms by decomposing $\widehat{Z}_{\mathcal{A}}^c$ via the Roelcke–Selberg spectral decomposition on $\mathcal{M}_{1,1}$. Each holomorphic eigenform of weight k contributes a critical line at $\text{Re}(s) = k/2$; Maass cusp-form projections contribute no new critical lines. $\square$

Example 3.12 (Standard families). The shadow metric data for the standard families:

Family	κ	α	S_4	Δ	Class	ρ
Heisenberg $\mathcal{H}_k$	k	0	0	0	$\mathbf{G}$	0
Affine $\widehat{\mathfrak{g}}_k$	$\frac{\dim \mathfrak{g}(k+h^\vee)}{2h^\vee}$	$\neq 0$	0	0	$\mathbf{L}$	0
$\beta\gamma$ (weight λ)	$6\lambda^2 - 6\lambda + 1$	0^*	0^*	0^*	$\mathbf{C}$	0
Virasoro Vir_c	$c/2$	2	$\frac{10}{c(5c+22)}$	$\frac{40}{5c+22}$	$\mathbf{M}$	$\frac{1}{c} \sqrt{\frac{180c+872}{5c+22}}$

* The table reports the weight-changing primary line, where $\kappa = S_3 = S_4 = 0$. The quartic contact invariant $Q^{\text{contact}} \neq 0$ lives on this charged stratum; the shadow metric requires $\kappa \neq 0$ and does not apply. The class **C** designation ($r_{\max} = 4$) comes from rank-one rigidity, not the single-line dichotomy. The T-line (stress tensor) carries the Virasoro shadow data $(\kappa, \alpha, S_4) = (c/2, 2, 10/(c(5c+22)))$ with $c = 2(6\lambda^2 - 6\lambda + 1)$ and is class **M**.

For the Virasoro algebra, the shadow metric evaluates to

$$Q_{\text{Vir}}(t) = c^2 + 12ct + \frac{180c + 872}{5c + 22} t^2,$$

with Gaussian decomposition $Q_{\text{Vir}} = (c + 6t)^2 + \frac{80}{5c+22} t^2$. The critical central charge c^* where $\rho = 1$ is the unique positive real root of

$$5c^3 + 22c^2 - 180c - 872 = 0,$$

approximately $c^* \approx 6.12$. For $c > c^*$ the shadow tower converges; for $c < c^*$ it diverges. At the Koszul self-dual point $c = 13$: $\rho(\text{Vir}_{13}) \approx 0.467$.

Remark 3.13 (Large- c asymptotic of S_4). The Virasoro quartic shadow

$$S_4(\text{Vir}_c) = \frac{10}{c(5c + 22)}$$

has large- c asymptotic

$$S_4(\text{Vir}_c) = \frac{10}{5c^2} \left(1 - \frac{22}{5c} + O(c^{-2}) \right) = \frac{2}{c^2} + O(c^{-3}),$$

not $2/(5c^2)$; the factor of 5 in the denominator of $10/[c(5c + 22)]$ is absorbed when expanding $1/(5c + 22)$ at large c . Pairing this with $\kappa = c/2$ gives the critical discriminant $\Delta = 8\kappa S_4 = 40/(5c + 22) \rightarrow 8/c$ as $c \rightarrow \infty$, confirming class **M** mixing persists to leading order.

The identity $S_2(\text{Vir}_c) = \kappa = c/2$ is convention-independent. The occasional error $S_2 = c/12$ conflates the shadow invariant with the Vol II lambda-bracket divided-power coefficient $c/12 = (c/2)/3!$, which is a different normalisation; the Vol I–II bridge is $S_2^{\text{Vol I}} = 6 S_2^\lambda$.

The Riccati seed on Vir_c is $(r_0, S_{r_0}) = (3, 2)$: $S_3(\text{Vir}_c) = 2$ is the c -independent cubic shadow (Lorgat, *The Virasoro spectral R-matrix*), and the shadow-exponential base $C_A = 6$ on the non-logarithmic class **M** locus factorises as $6 = r_0 \cdot S_{r_0} = 3 \cdot 2$.

Remark 3.14 (Spectral curve). The algebraic relation $H^2 = t^4 Q_L(t)$ defines, for each primary line L , a spectral curve $\Sigma_L := \{H^2 = t^4 Q_L(t)\} \subset \mathbb{A}_{t,H}^2$. Since Q_L is quadratic in t , Σ_L has arithmetic genus 0; the projection $(t, H) \mapsto t$ is a double cover ramified at the zeros of Q_L . The degeneration loci in parameter space are controlled by the critical discriminant: Σ_L acquires a node precisely when $\Delta = 0$. The shadow growth rate ρ_L is the reciprocal of the distance from $t = 0$ to the nearest ramification point.

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